

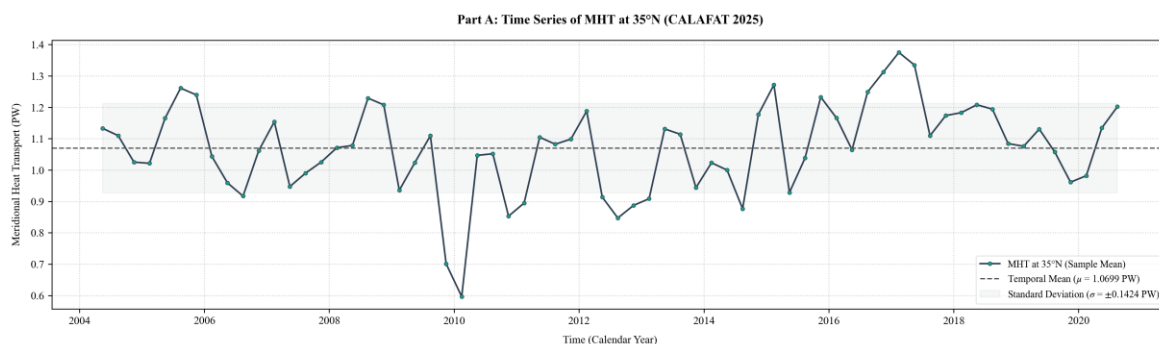
Assignment 1 — Characterising an AMOC time series

1 Data

- selected dataset: The Calafat 2025 Reconstructed AMOC dataset
- target variable: Meridional Heat Transport(MHT), unit: Petawatts(PW)
- location: 35°N(from 30.5°N to 40.0°N)
- time span: 2004-2-14 to 2020-8-14

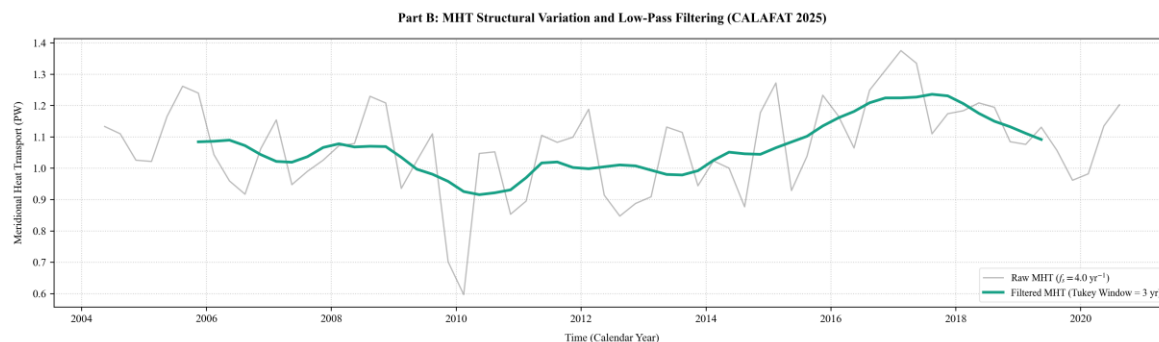
2 Progress

2.1 Part A

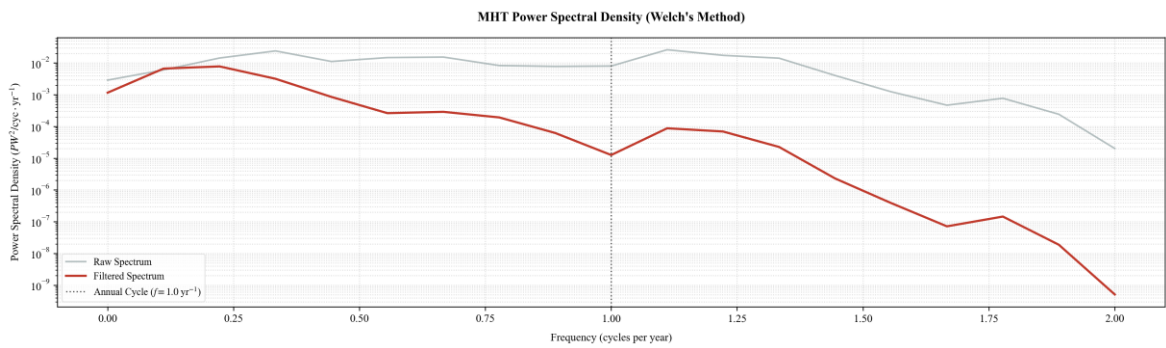


- This figure shows the full time series of Meridional Heat Transport at 35 degrees North from 2004 to 2020. It includes the average value line and a shaded area representing one standard deviation to show the regular up and down changes over time.

2.2 Part B

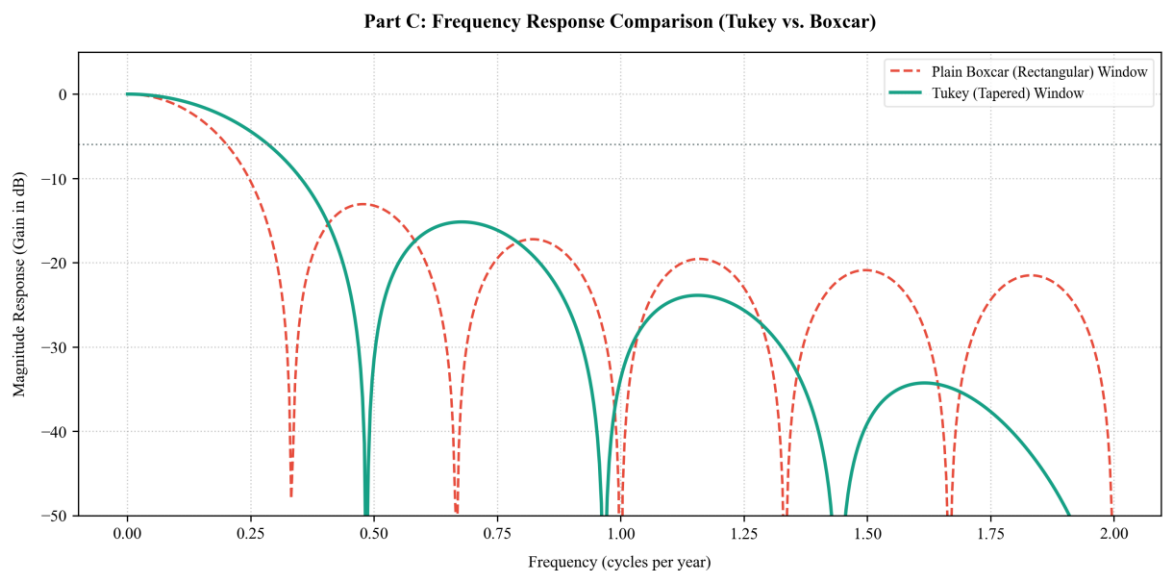


- This plot compares the original data with a smoothed line created by a 3-year Tukey filter. This filter removes the quick seasonal changes to show the longer climate trends clearly.



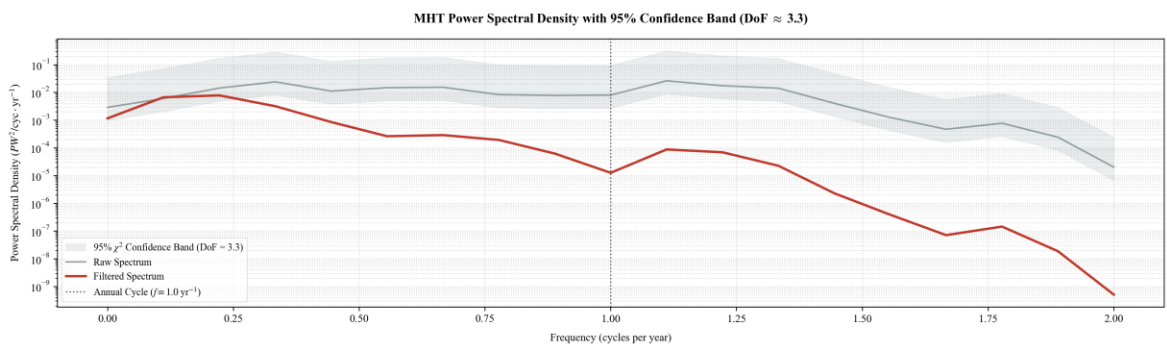
- This graph shows the energy of the data at different frequencies using Welch's method. It proves that the filter, shown by the red line, successfully reduces the power of fast changes, with a dotted line marking the 1-year cycle.

2.3 Part C



- This graph shows that the Tukey filter drops off much faster at high frequencies than the Boxcar filter, making it better at blocking fast background noise.

2.4 Confidence Intervals



- This graph adds a light grey 95% confidence band to show statistical error, showing that the filtered red line drops completely below this band.